



IoT: Building Intelligent Systems

The Technology Behind Smart and Connected Environments



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Connected

Devices and systems work together



Smart

Real-time data drives intelligence



Scalable

Cloud platforms enable growth



Secure

Protection of data and devices



Efficient

Automation improves performance





Core Concepts of IoT

IoT is a network of connected physical devices that collect, exchange, and process data using sensors, networks, and cloud platforms.



Sensors

Collects real-world data



Network

Transfers data via wireless links



Cloud

Stores and processes data



Automation

Enables smart responses

Layers of an IoT Ecosystem

IoT systems are organized into three functional layers, each responsible for a distinct role in the end-to-end data pipeline.



Device Layer

Sensors, actuators, and microcontrollers, such as Arduino or Raspberry Pi, form the point of data origin.



Network Layer

Wi-Fi, Bluetooth, Zigbee, and MQTT route data between devices and the cloud.



Cloud Layer

Cloud platforms like AWS IoT and Azure IoT Hub store, analyze data, and issue control commands.



Practical Applications of IoT



Smart Parking System



1. Sensors detect empty parking spots



2. Data sent to cloud over Wi-Fi / 4G



3. Users find & reserve spots via mobile app



4. Automated entry & exit with digital payment



Smart Waste Management



1. Sensors monitor bin fill levels



2. Data sent to cloud in real time



3. Routes optimized for collection trucks



4. Reduces fuel use & keeps cities clean

IoT Communication Workflow

This complete IoT communication cycle turns physical sensing into automated action.



IoT Across Modern Sectors

IoT connects devices, collects data, and enables smarter decisions across diverse domains.



Smart Agriculture

- Soil moisture & nutrient monitoring
- Livestock tracking
- Crop health monitoring



Smart Transportation

- Real-time vehicle tracking
- Predictive maintenance
- Traffic flow optimization



Smart Building

- Energy usage monitoring
- HVAC optimization
- Security & access control



Environmental Monitoring

- Air quality monitoring
- Water quality tracking
- Weather condition monitoring



Why It Matters

IoT solutions improve efficiency, reduce costs, enhance safety, and create a better, sustainable future.

Connected today, smarter tomorrow.



Agriculture



Transportation



Buildings



Environment



Healthcare



Manufacturing



Education



Public Safety



Retail

Core Design of IoT Systems

Every IoT system — regardless of domain — follows the same core design principle:

Input → Processing → Output



IoT systems collect input from sensors, process data locally or in the cloud, and trigger outputs as actions in the physical world.



Sensors & Actuators

Interface between the physical world and the digital system.



Controllers (MCU/MPU)

Local logic execution before data is transmitted upstream.



Reliable Communication

Protocol selection (MQTT, HTTP, CoAP) governs latency and reliability.



Efficient Design

Power consumption, bandwidth, and security must be balanced.



Sensors capture signals, controllers interpret them, and actuators turn decisions into real-world action.



Conclusion

IoT powers smart, connected systems that transform the way we live and work.



Smart Automation

Machines sense, decide, and act with less manual effort.



Real-Time Monitoring

Live data improves efficiency, safety, and resource use.



Future Scope

IoT will continue to drive innovation in smart homes, healthcare, cities, and Industry 4.0.



Improved Efficiency

Automate tasks and reduce human effort.



Enhanced Safety

Real-time alerts and risk prevention.



Optimized Resources

Better utilization and lower operational cost.



Smarter Decisions

Data-driven insights for better outcomes.

